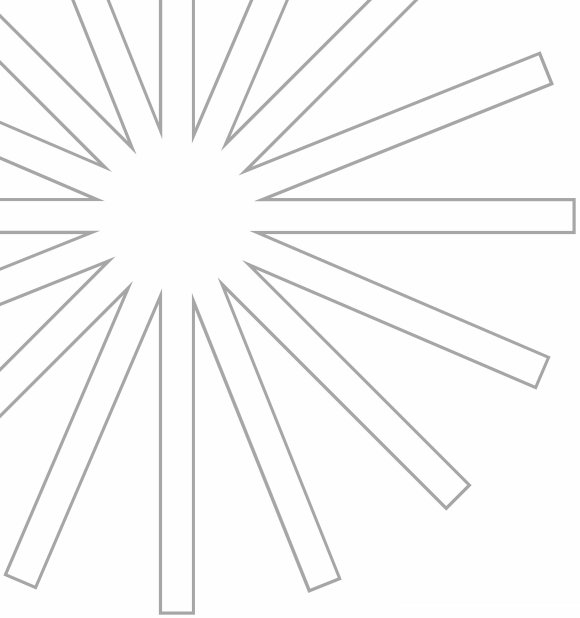
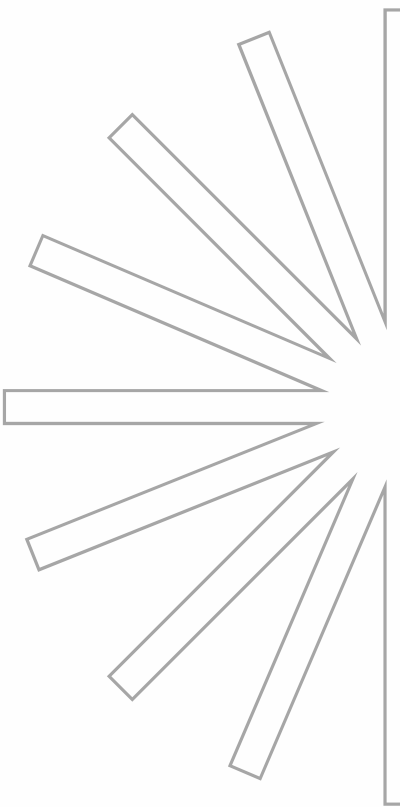




DEVELOPMENT OF A DIGITAL FORENSIC ANALYSIS SYSTEM FOR MOBILE APPLICATIONS

6B06301 – Cybersecurity
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PROBLEM

- Mobile applications store sensitive forensic artifacts.
- Mobile incidents generate large volumes of digital evidence.
- Existing forensic tools require extensive manual analysis.
- Manual investigations are slow and prone to missed indicators.

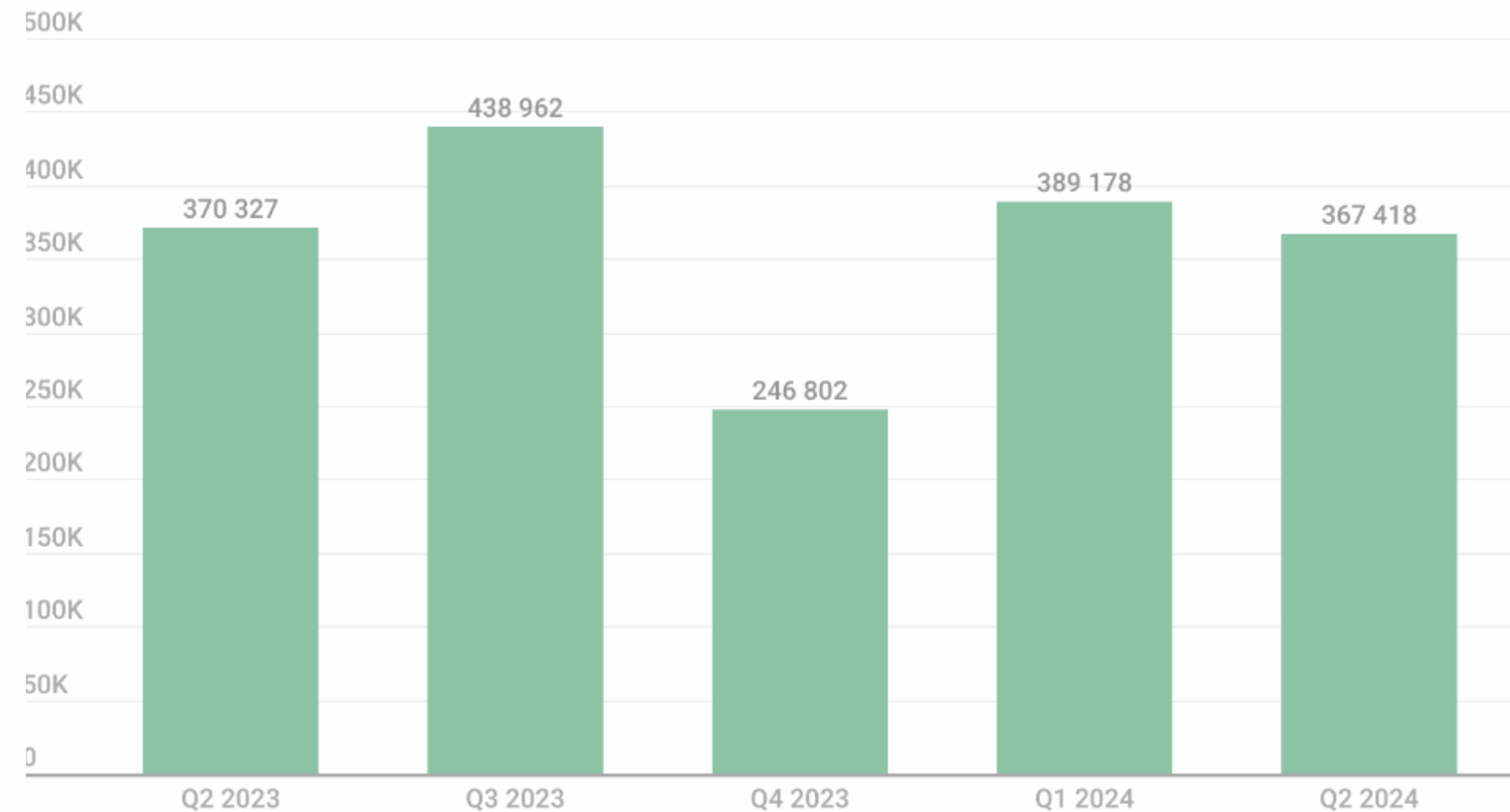


Figure: Statistics of cybersecurity incidents related to mobile threats

Source: Kaspersky Cybersecurity Report, 2024

kaspersky

Need for Automated Mobile Forensic Analysis

LITERATURE REVIEW

Source	Key Finding
NIST SP 800-101	Standard forensic methodology
Casey (2019)	Reliable evidence handling
Tamma et al. (2020)	Mobile artifacts as evidence
Kumar & Shukla (2022)	Artifact extraction focus
Abraham & Nair (2023)	AI-assisted detection

Total analyzed sources: 40+ (2020–2025)

GOAL & OBJECTIVES

Develop an automated digital forensic analysis system for detecting, analyzing and reporting security-related artifacts in mobile applications.

Analyze Existing Tools

- Review current forensic solutions
- Identify limitations

Identify Mobile Artifacts

- Determine security artifacts
- Define evidence sources

Design System Architecture

- Create analysis workflow
- Define system modules

Develop & Evaluate Prototype

- Implement core functions
- Assess effectiveness

Based on NIST guidelines (mobile device forensics)

METHODOLOGY

Analytical Method	Engineering Method	Experimental Method	Comparative Method
<i>Analysis of existing mobile forensic research, NIST SP 800-101 guidelines, and forensic tools.</i>	<i>Design and implementation of a modular Python-based forensic platform.</i>	<i>Testing on 157 labelled files across 4 forensic datasets.</i>	<i>Comparison with Cellebrite UFED, Oxygen Forensic Detective, and MobSF using key forensic criteria.</i>
Output → Research gap identification	Output → Working prototype	Output → Comparative analysis	Output → Comparative evaluation

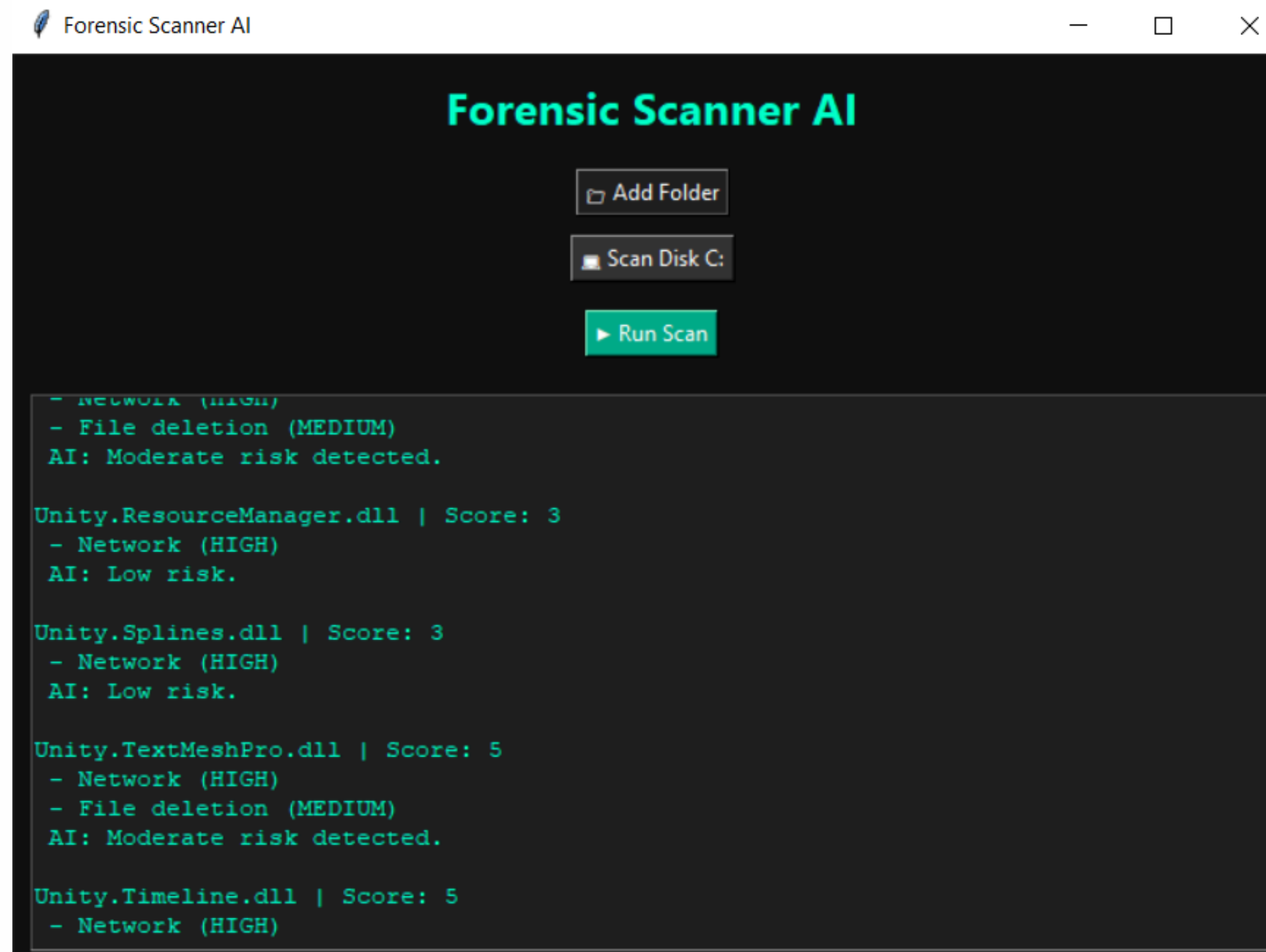
COMPARATIVE ANALYSIS

Feature	Cellebrite	Oxygen	MobSF	Proposed System
Automation	Low	Medium	Medium	High
Artifact Correlation	Limited	Limited	No	Yes
Explainability	Medium	Medium	High	High
App-Level Analysis	Low	Medium	High	High

The proposed system provides the highest level of automation, artifact correlation, and explainable analysis while maintaining strong application-level forensic capabilities.

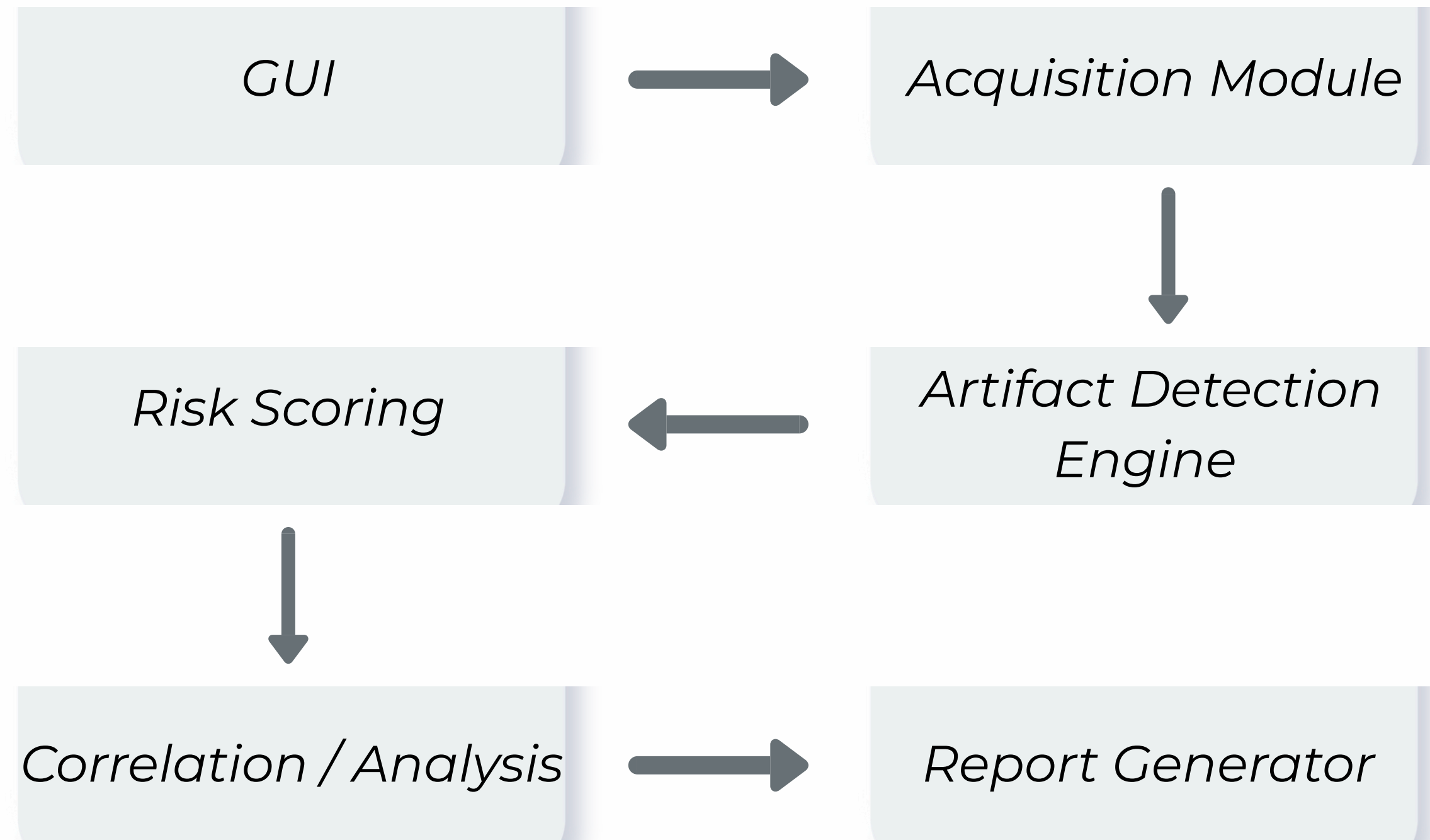
FORENSIC ANALYSIS WORKFLOW

Automated workflow from artifact acquisition to forensic reporting.



Prototype interface during forensic artifact analysis

SYSTEM ARCHITECTURE



Modular architecture of the proposed forensic analysis system.

FUTURE WORK

Current System



Rule-Based Analysis



Machine Learning
Support

Anomaly Detection Models

Isolation Forest for automated anomaly detection in forensic artifacts.

Artifact Classification

Random Forest for classification of suspicious and benign artifacts.

Behavioral Pattern Recognition

Detection of recurring suspicious activity patterns across applications.

Accuracy Improvement

Enhancement of detection performance using data-driven approaches.

RISK SCORING MODEL

Artifact	Severity	Score
Token	Critical	5
IP Address	High	3
Deletion Trace	Medium	2
Authentication Record	Low	1

- $\text{Score} \geq 8 \rightarrow \text{High Risk}$
- $\text{Score} \geq 5 \rightarrow \text{Moderate Risk}$
- $\text{Score} < 5 \rightarrow \text{Low Risk}$

```
def calculate_score(findings):
    score = 0

    for finding in findings:
        severity = finding[1]

        if severity == "CRITICAL":
            score += 5

        elif severity == "HIGH":
            score += 3

        elif severity == "MEDIUM":
            score += 2

        elif severity == "LOW":
            score += 1

    return score

def ai_analysis(findings):
    score = calculate_score(findings)

    if score >= 8:
        return "High likelihood of compromise detected"

    elif score >= 5:
        return "Moderate risk detected"

    else:
        return "Low risk"
```

Figure: Risk scoring logic for forensic artifact evaluation

Source: Author's implementation

PROTOTYPE

Implemented Components

- GUI-based desktop application
- Artifact detection engine
- Risk scoring module
- Automated report generator

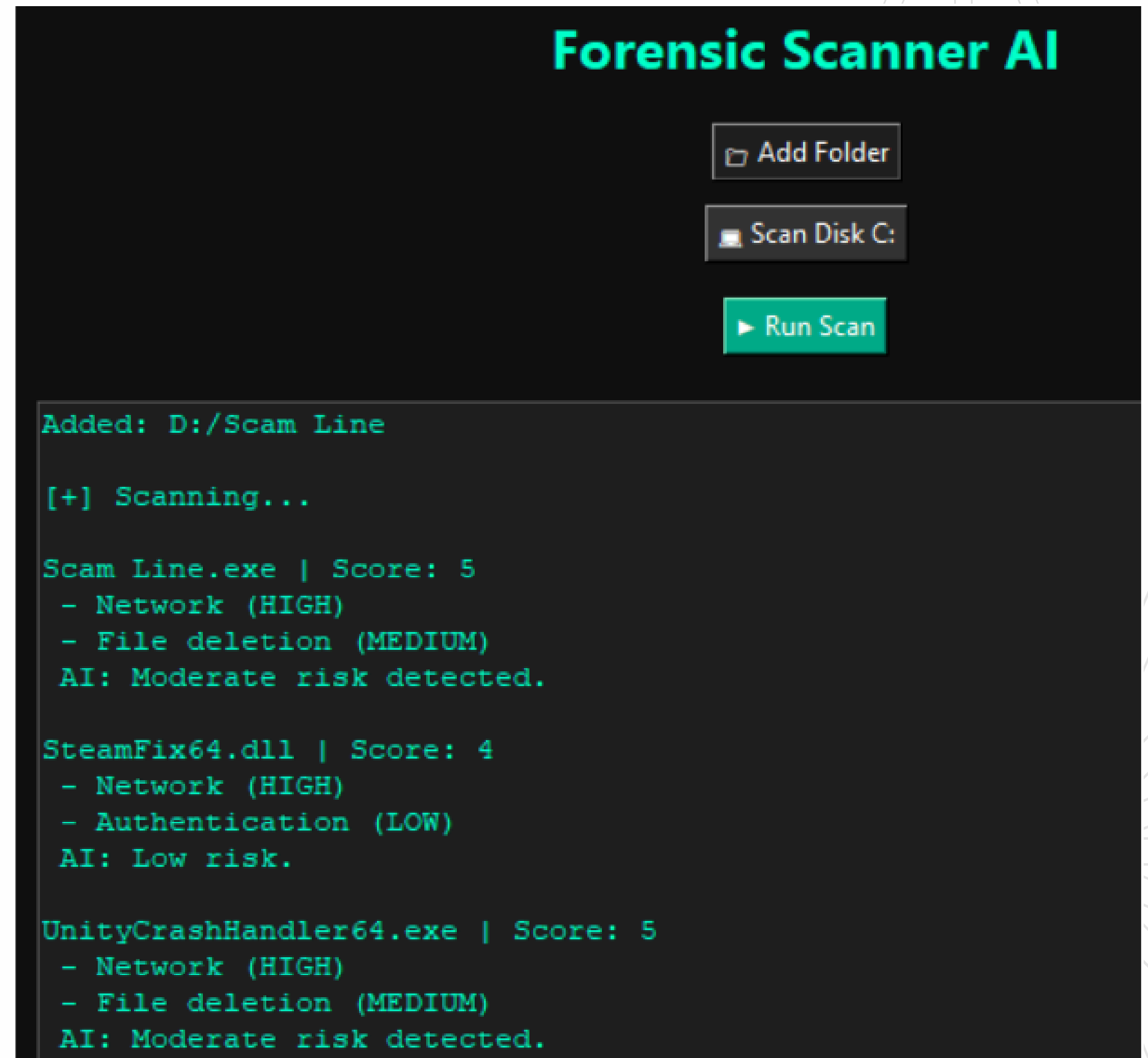
Supported Artifacts

- Authentication tokens
- IP addresses
- Log files
- File deletion traces

Output Formats

- Analysis report
- Risk assessment summary
- Court-ready PDF report

Figure: Main graphical interface of the developed forensic system
Source: Author's implementation



Functional prototype implementing artifact detection, risk scoring, and automated forensic reporting.

RESULTS

Artifact	Severity
Token	Critical
IP Address	High
File Deletion	Medium

Example Assessment
Total Risk Score: 10
System Verdict: High likelihood of compromise

File: settings.json
Path: D:/Scam Line/Scam Line_Data/StreamingAssets\aa\settings.json
Risk Score: 3
- Network (HIGH)
AI Analysis: Low risk. No strong indicators of compromise.

File: link.xml
Path: D:/Scam Line/Scam Line_Data/StreamingAssets\aa\AddressablesLink\link.xml
Risk Score: 3
- Network (HIGH)
AI Analysis: Low risk. No strong indicators of compromise.

Risk Analysis Chart

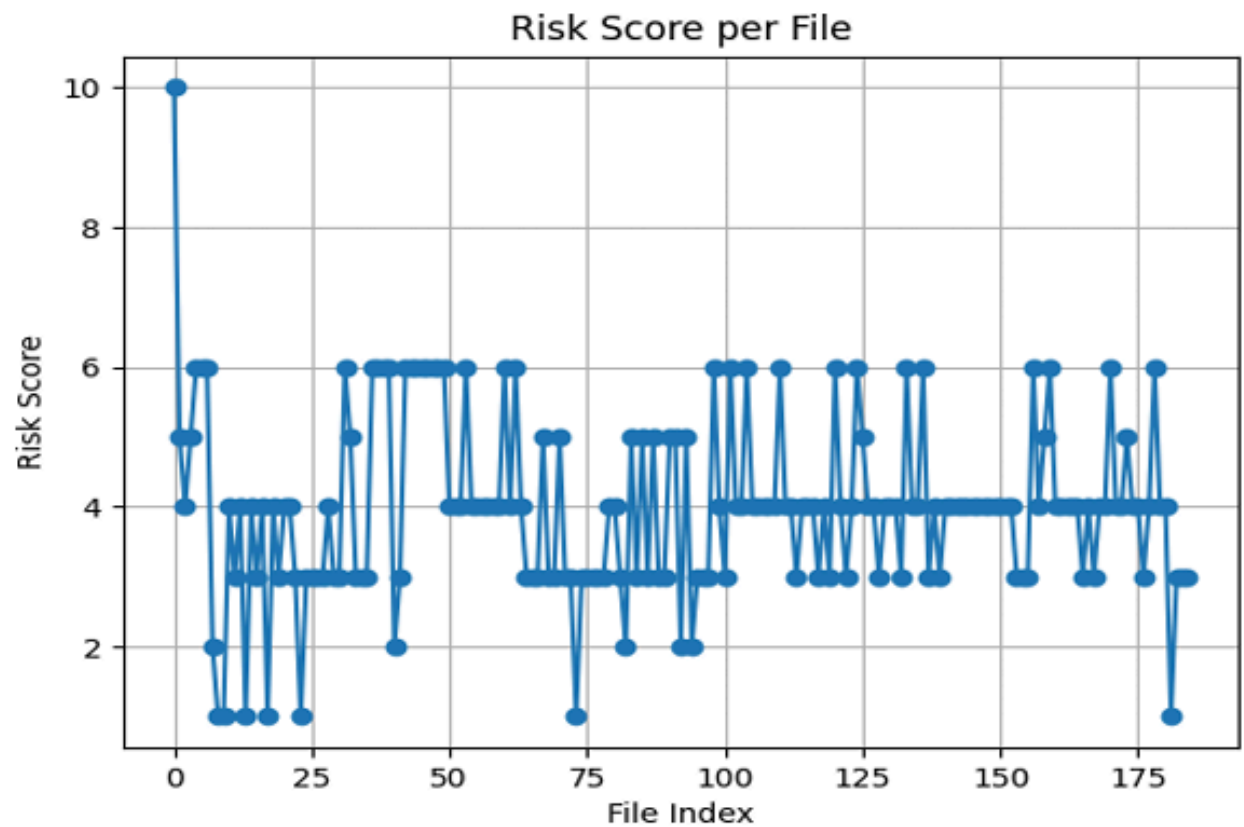


Figure: Generated forensic analysis results and risk visualization
Source: Generated during experimental testing

AUTOMATED REPORT GENERATION

```
COURT_REPORT_8.txt – Блокнот
Файл  Правка  Формат  Вид  Справка
DIGITAL FORENSIC EXAMINATION REPORT
=====

Anomaly 1:
Type: Network
Severity: HIGH
File: Scam Line.exe
Path: D:/Scam Line\Scam Line.exe
Hash: 46ee0ae2a675d27c7b1728570282b9e1

Anomaly 2:
Type: File deletion
Severity: MEDIUM
File: Scam Line.exe
Path: D:/Scam Line\Scam Line.exe
Hash: 46ee0ae2a675d27c7b1728570282b9e1

Anomaly 3:
Type: Network
Severity: HIGH
File: SteamFix64.dll
Path: D:/Scam Line\SteamFix64.dll
Hash: a9b5b64734caaa251ad55d9de46a958b
```

Anomaly 8:
Type: File deletion
Severity: MEDIUM
File: cudart64_75.dll
Path: C:/rym\cudart64_75.dll
Hash: 60ea51e513ffc81a2786b9cca098cab7

Anomaly 9:
Type: Authentication
Severity: LOW
File: cudart64_75.dll
Path: C:/rym\cudart64_75.dll
Hash: 60ea51e513ffc81a2786b9cca098cab7

Anomaly 10:
Type: File deletion
Severity: MEDIUM
File: cudart64_80.dll
Path: C:/rym\cudart64_80.dll
Hash: cf198b329fb988983749f891c060245e

EVALUATION METRICS

EXPERIMENTAL EVALUATION

- 4 TEST DATASETS
- 157 ANALYZED FILES
- ANALYSIS TIME < 1 MINUTE PER DATASET

PERFORMANCE METRICS

- PRECISION: 87.5%
- RECALL: 80.8%
- ACCURACY: 94.9%

CONFUSION MATRIX

- $TP = 21$
- $FP = 3$
- $TN = 128$
- $FN = 5$

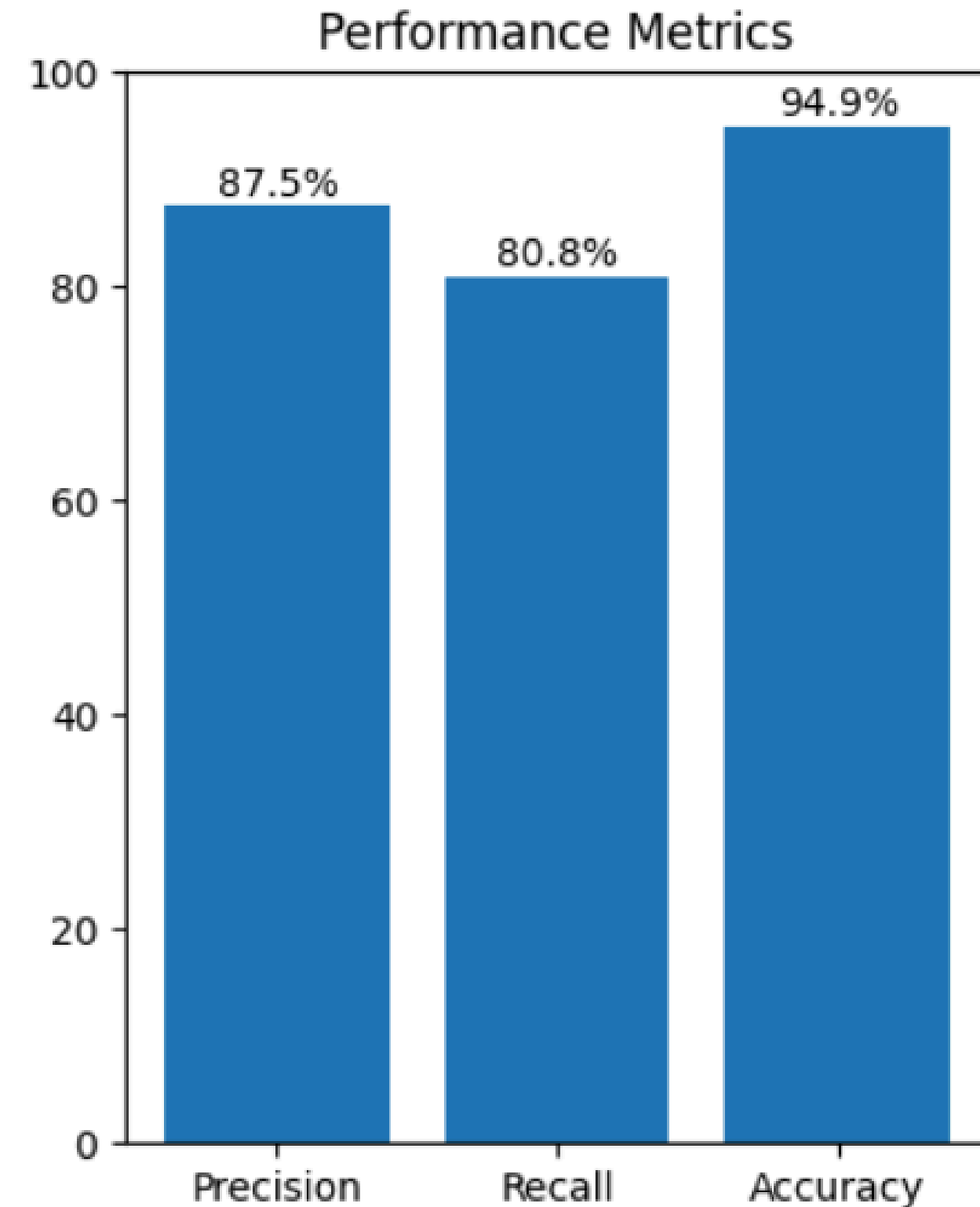


Figure: Experimental evaluation metrics of the proposed system
Source: Generated during testing

SYSTEM ADVANTAGES

Rule-Based Detection
Explainable Results

Risk Scoring
Prioritized Investigation

Automated Reports
Reduced Manual Work

94.9%
Accuracy

< 1 Minute
Analysis

157 Files Tested

CONCLUSION

Achieved Results

- *Developed a working forensic analysis prototype*
- *Detected security-related artifacts (tokens, IP addresses, file deletion traces)*
- *Implemented rule-based risk assessment and explainable analysis*
- *Generated automated forensic reports*

Experimental Results

- *Accuracy: 94.9%*
- *Analysis time: < 1 minute per dataset*
- *Evaluated on 157 files across 4 datasets*

Final Outcome

The proposed system automates key stages of mobile forensic analysis and provides explainable investigation results.

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THANK
YOU!

